

# MATTHEW SANTALI GANA

Full-Stack Software Engineer | Backend Engineer | Frontend Engineer | Software Architect

**Address:** Abuja, Nigeria | Open to Remote, Hybrid, On-site & Relocation Opportunities |

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**GitHub:** github.com/matthewgana | **Portfolio:** https://gmatts-authority-platform.vercel.app

## PROFESSIONAL SUMMARY

Full-Stack Software Engineer with expertise in designing and building secure, scalable, and maintainable software systems across frontend and backend environments. Experienced in developing production-ready web applications using React, Next.js, TypeScript, Node.js, NestJS, Express.js, PostgreSQL, MongoDB, Redis, and Prisma. Strong background in software architecture, API design, authentication and authorization, application security, database design, and modern engineering practices. Passionate about building intelligent platforms that solve real-world problems across business, education, civic technology, and enterprise software. Combines an engineering mindset with a background in Physics & Electronics, STEM education, cybersecurity, technical writing, and AI-assisted software development to deliver reliable, user-focused software solutions.

## CORE TECHNICAL SKILLS

- **Languages:** TypeScript • JavaScript • Python • SQL
- **Frontend:** React • Next.js • HTML5 • CSS3 • CSS Modules • Responsive Design • Accessibility (WCAG/ARIA) • Server-Side Rendering (SSR) • Static Site Generation (SSG)
- **Backend:** Node.js • NestJS • Express.js • REST APIs • JWT Authentication • Refresh Token Rotation • Role-Based Access Control (RBAC) • Attribute-Based Access Control (ABAC)
- **Data, Performance & Infrastructure:** PostgreSQL, MongoDB, Redis, BullMQ, API Optimization, Caching Strategies, Background Job Processing
- **Analytics & Tooling:** Python • Pandas • NumPy • Power BI • Advanced Excel • Data Cleaning • ETL • Data Visualization
- **DevOps & Deployment:** Git • GitHub • CI/CD • Vercel • Render • Environment Management • Testing Strategies • Modular Architecture
- **Security:** OWASP Principles • API Validation • Rate Limiting • Secure Authentication • Authorization • Zero Trust Concepts

## SELECTED SOFTWARE ENGINEERING PROJECTS

### FOUNDRA — AI-Native Business Operating System (In Development)

*Role: Founder & Full-Stack Software Engineer*

*Technology Stack: React • Next.js • TypeScript • Node.js • NestJS • PostgreSQL • Prisma • AI Integration*

- Architecting an AI-native business operating system that supports entrepreneurs from business idea Designed and developed an AI-native business operating system that supports entrepreneurs from business planning through day-to-day operations and business growth.
- Built a modular full-stack architecture for customer management, inventory, finance, analytics, and AI-assisted business workflows.

- Engineered scalable backend services and reusable APIs with a focus on maintainability, security, and long-term extensibility.
- Implemented responsive frontend interfaces using React, Next.js, and TypeScript to deliver an intuitive user experience.
- Currently refining the platform through feature enhancements, performance optimization, and production readiness.

## **R-AILOS — Resilient AI Learning & Safety Operating System (In Development)**

***Role: Founder & Full-Stack Software Engineer***

***Technology Stack: React • Next.js • TypeScript • Node.js • NestJS • PostgreSQL • AI Integration***

- Designed and developed an offline-first, AI-native platform that improves access to quality STEM education in low-connectivity environments.
- Engineered modular frontend and backend systems for learning management, multilingual content delivery, user management, and AI-assisted learning experiences.
- Built scalable APIs and secure application architecture to support long-term platform growth and maintainability.
- Applied modern software engineering practices to create a resilient, secure, and extensible educational platform.
- Currently iterating on platform capabilities, performance, and production readiness.

## **AgriTrack — Agricultural Input Cost Tracker & Cooperative Buying Platform**

***Role: Lead Full-Stack Software Engineer & Mobile Developer***

***Technology Stack: React • React Native • Next.js • TypeScript • NestJS • PostgreSQL • MongoDB • Prisma • CSS Modules • GSAP***

- Designed and developed a full-stack agricultural management platform that enables farmers to track input costs, compare supplier prices, and support cooperative purchasing.
- Led the end-to-end software development lifecycle, including system architecture, UI implementation, backend development, database design, API development, testing, deployment, and ongoing platform enhancement.
- Refactored the backend from Express.js to NestJS to improve modularity, maintainability, scalability, and long-term extensibility.
- Designed and implemented secure RESTful APIs, authentication, authorization, input validation, and role-based access control using modern backend engineering practices.
- Engineered a hybrid data architecture using PostgreSQL for relational data and MongoDB for flexible document storage to support evolving business requirements.
- Built responsive, accessible, and high-performance user interfaces using React, Next.js, TypeScript, CSS Modules, and GSAP.
- Deployed the web application using Vercel, Render, and GitHub-based development workflows.
- Currently developing a cross-platform mobile application using React Native to extend the platform across Android and iOS devices.

## **Personal Developer Portfolio Platform**

***Role: Full-Stack Software Engineer***

***Technology Stack: Next.js • React • TypeScript • CSS Modules • Framer Motion***

- Designed and developed a production-ready portfolio platform showcasing software engineering projects and technical expertise.

- Built a server-first architecture using Server-Side Rendering (SSR) and Static Site Generation (SSG) to improve performance and SEO.
- Developed reusable, strongly typed UI components and a scalable frontend architecture.
- Implemented accessibility standards, responsive design, structured metadata, and JSON-LD to improve user experience and search visibility.
- Deployed the platform on Vercel using GitHub-based continuous deployment workflows.

## PROFESSIONAL EXPERIENCE

### Full-Stack Software Engineer (Capstone Project)

#### TechCrush Cohort 6 | 2026

- Collaborated with product managers, product designers, frontend developers, backend developers, mobile application developers, and technical writers to deliver a full-stack software solution using Agile development practices.
- Led the backend engineering track, coordinating implementation efforts, reviewing technical decisions, and supporting teammates throughout the development lifecycle.
- Designed and implemented secure RESTful APIs, authentication, authorization, and database integration using modern backend engineering practices.
- Worked closely with frontend developers to define API contracts, improve system integration, and deliver a seamless user experience.
- Participated in architecture discussions, sprint planning, code reviews, testing, debugging, and deployment activities while maintaining engineering best practices.

### Mathematics Teacher & STEM Content Developer

#### 2019 – Present

- Teach Mathematics, Further Mathematics, Physics, and Chemistry to secondary school students, helping learners develop strong analytical and problem-solving skills.
- Design lesson plans, assessments, and learning resources that simplify complex STEM concepts and improve student understanding.
- Develop written STEM educational resources, including ebooks, study guides, and technical learning materials.
- Author technical manuscripts covering software engineering, backend development, cybersecurity, artificial intelligence, data analytics, and web development.
- Currently expanding these written resources into broader digital learning content.

## CERTIFICATIONS

- **Backend Development** — TechCrush for Africans (2026)
- **Cybersecurity** — TechCrush for Africans (2025)
- **Data Analytics** — Xellux Technologies Institute (2025)
- **Artificial Intelligence & Machine Learning** — In Progress (Expected Completion: September/October 2026)

## EDUCATION

- **HND, Physics / Electronics (Distinction)** — Federal Polytechnic, Bida
- **ND, Science Laboratory Technology (Upper Credit)** — Federal Polytechnic, Bida